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JEE MAIN PHYSICS DPP: APPLICATIONS OF DIMENSIONAL ANALYSIS

Topic: Homogeneity, Checking Correctness & Deriving Physical Relations | Time: 45 Min | Max Marks: 80

Marking Scheme: +4, -1 | Designed & Curated by: Syed Mazhar Ali

Q1. If velocity v , force F , and time T are chosen as the fundamental base quantities, the dimensional formula of mass m is:

- (a) $[F^1 v^{-1} T^1]$
- (b) $[F^1 v^1 T^{-1}]$
- (c) $[F^1 v^{-2} T^1]$
- (d) $[F^1 v^{-1} T^2]$

Q2. The period of oscillation T of a gas bubble from an underwater explosion depends upon static pressure P , density of water ρ , and total energy of explosion E as $T \propto P^a \rho^b E^c$. The values of a, b , and c are respectively:

- (a) $-\frac{5}{6}, \frac{1}{2}, \frac{1}{3}$
- (b) $-\frac{5}{6}, \frac{1}{3}, \frac{1}{2}$
- (c) $\frac{5}{6}, -\frac{1}{2}, \frac{1}{3}$
- (d) $-\frac{1}{2}, \frac{1}{3}, \frac{5}{6}$

Q3. In the equation $y = A \sin(\omega t - kx)$, the dimensional formula of ω/k represents:

- (a) Velocity
- (b) Acceleration
- (c) Wavelength
- (d) Time period

Q4. If energy E , velocity v , and force F are taken as fundamental quantities, the dimensional formula of surface tension T is:

- (a) $[E^{-2} v^0 F^2]$
- (b) $[E^{-1} v^0 F^2]$
- (c) $[E^1 v^{-2} F^2]$
- (d) $[E^{-2} v^2 F^1]$

Q5. The velocity of water waves v depends on their wavelength λ , density of water ρ , and acceleration due to gravity g as $v \propto \lambda^a g^b \rho^c$. Dimensional analysis confirms that:

- (a) $a = 1/2, b = 1/2, c = 0$
- (b) $a = 1, b = 1/2, c = 0$
- (c) $a = 1/2, b = 1, c = 1$
- (d) $a = 1/2, b = -1/2, c = 0$

Q6. Which of the following relationships is dimensionally inconsistent?

- (a) $v^2 = u^2 + 2as$

(b) $s_n = u + \frac{a}{2}(2n - 1)$

(c) $T = 2\pi \sqrt{\frac{m}{k}}$ (spring-mass system)

(d) $\lambda = \frac{h}{mv^2}$ (de-Broglie wavelength)

Q7. The force of interaction between two atoms is modeled as $F = \alpha \beta \exp\left(-\frac{x^2}{\alpha k_B T}\right)$, where x is distance, k_B is Boltzmann's constant, and T is absolute temperature. The dimensional formula of β is:

- (a) $[M^1 L^1 T^{-2}]$
- (b) $[M^2 L^2 T^{-4}]$
- (c) $[M^1 L^{-1} T^{-2}]$
- (d) $[M^0 L^1 T^{-1}]$

Q8. If Planck's constant h , speed of light c , and gravitational constant G are chosen as fundamental units, the dimension of length in this system is:

- (a) $[h^{1/2} G^{1/2} c^{-3/2}]$
- (b) $[h^{1/2} G^{1/2} c^{3/2}]$
- (c) $[h^{-1/2} G^{1/2} c^{3/2}]$
- (d) $[h^{1/2} G^{-1/2} c^{-3/2}]$

Q9. If momentum p , area A , and time t are taken as fundamental units, the dimensional formula for energy is:

- (a) $[p^1 A^{1/2} t^{-1}]$
- (b) $[p^1 A^1 t^{-1}]$
- (c) $[p^2 A^{1/2} t^{-1}]$
- (d) $[p^1 A^{-1/2} t^1]$

Q10. The critical velocity v_c of a viscous liquid flowing through a capillary tube of diameter D depends on viscosity η , density ρ , and D . The dimensionally correct relation is:

- (a) $v_c = \frac{R_e \eta}{\rho D}$
- (b) $v_c = \frac{R_e \rho D}{\eta}$
- (c) $v_c = \frac{R_e \rho}{\eta D}$
- (d) $v_c = \frac{R_e \eta D}{\rho}$

Q11. In the equation $\left(P + \frac{\alpha}{V^2}\right) = \frac{b\theta}{V} \exp\left(-\frac{\alpha V}{R\theta}\right)$, where P is pressure, V is volume, θ is temperature, and R is universal gas constant, the dimension of α must be:

- (a) $[M^1 L^5 T^{-2}]$

- (b) $[M^1 L^2 T^{-2}]$
 (c) $[M^0 L^{-3} T^0]$
 (d) $[M^1 L^{-1} T^{-2}]$
- Q12.** If Planck's constant h , speed of light c , and universal gravitational constant G are chosen as fundamental units, the Planck mass m_P scales as:
- (a) $[h^{1/2} c^{1/2} G^{-1/2}]$
 (b) $[h^{1/2} c^{-1/2} G^{1/2}]$
 (c) $[h^{-1/2} c^{1/2} G^{1/2}]$
 (d) $[h^1 c^1 G^{-1}]$
- Q13.** A dimensionless combination formed by electron charge e , permittivity of free space ϵ_0 , Planck's constant h , and speed of light c is:
- (a) $\frac{e^2}{\epsilon_0 h c}$
 (b) $\frac{e^2 \epsilon_0}{h c}$
 (c) $\frac{e h}{\epsilon_0 c^2}$
 (d) $\frac{e^2 c}{\epsilon_0 h}$
- Q14.** The rate of flow of liquid V/t (Q) through a pipe of radius r under pressure gradient P/l depends on coefficient of viscosity η as $Q \propto \eta^a r^b (P/l)^c$. Dimensional analysis yields:
- (a) $a = -1, b = 4, c = 1$
 (b) $a = 1, b = 4, c = -1$
 (c) $a = -1, b = 2, c = 1$
 (d) $a = 1, b = 2, c = 1$
- Q15.** If force F , velocity v , and density ρ are fundamental quantities, the dimensional formula for electrical resistance R is:
- (a) Dependent on electric charge/current unit $[A]$
 (b) Undefinable using only $\{F, v, \rho\}$ without an electromagnetic base dimension
 (c) Both (a) and (b) are correct
 (d) $[F^1 v^{-1} \rho^0]$
- Q16.** In the relation $P = P_0 \exp(-\alpha t^2)$, where P is pressure and t is time, the dimensional formula of constant α is:
- (a) $[M^0 L^0 T^{-2}]$
 (b) $[M^1 L^{-1} T^{-2}]$
 (c) $[M^0 L^0 T^2]$
 (d) $[M^1 L^0 T^{-2}]$
- Q17.** The frequency of vibration ν of a stretched string depends on length L , tension F , and linear mass density μ (mass per unit length) as $\nu \propto L^a F^b \mu^c$. The values of a, b , and c are:
- (a) $-1, 1/2, -1/2$
- (b) $1, 1/2, 1/2$
 (c) $-1, -1/2, 1/2$
 (d) $-1/2, 1/2, -1$
- Q18.** A physical relation is given as $x = \frac{a}{\sqrt{2}} \sin\left(\frac{2\pi vt}{b}\right) \cos\left(\frac{2\pi x}{c}\right)$, where x is displacement, v is speed, and t is time. The dimensional formula of (ab/c) is:
- (a) $[M^0 L^1 T^0]$
 (b) $[M^0 L^1 T^{-1}]$
 (c) $[M^0 L^0 T^0]$
 (d) $[M^0 L^2 T^{-1}]$
- Q19.** The centripetal force F acting on a particle moving in a circle depends on mass m , velocity v , and radius r as $F = km^a v^b r^c$. The exponents are:
- (a) $a = 1, b = 2, c = -1$
 (b) $a = 1, b = 1, c = -2$
 (c) $a = 2, b = 1, c = -1$
 (d) $a = 1, b = 2, c = 1$
- Q20.** Which of the following is a major limitation of the method of dimensional analysis?
- (a) It cannot determine dimensionless proportionality constants.
 (b) It cannot derive relations containing trigonometric, exponential, or logarithmic functions.
 (c) It fails when a physical quantity depends on more than three mechanical variables (M, L, T).
 (d) All of the above.

ANSWER KEY & STEP-BY-STEP SOLUTIONS

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Quick Answer Key

Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans
1	(a)	6	(d)	11	(c)	16	(a)
2	(a)	7	(c)	12	(a)	17	(a)
3	(a)	8	(a)	13	(a)	18	(a)
4	(b)	9	(a)	14	(a)	19	(a)
5	(a)	10	(a)	15	(c)	20	(d)

Detailed Solutions

Sol 1. Correct Option: (a)

Let $m \propto F^a v^b T^c$.

$$[M^1 L^0 T^0] = [M^1 L^1 T^{-2}]^a [L^1 T^{-1}]^b [T^1]^c$$

Equating powers:

$$M : a = 1$$

$$L : a + b = 0 \Rightarrow b = -1$$

$$T : -2a - b + c = 0 \Rightarrow -2(1) - (-1) + c = 0 \Rightarrow c = 1$$

$$\text{Thus, } [m] = [F^1 v^{-1} T^1].$$

Sol 2. Correct Option: (a)

$$T \propto P^a \rho^b E^c$$

$$[T^1] = [ML^{-1}T^{-2}]^a [ML^{-3}]^b [ML^2T^{-2}]^c$$

$$M : a + b + c = 0$$

$$L : -a - 3b + 2c = 0$$

$$T : -2a - 2c = 1 \Rightarrow a + c = -1/2$$

$$\text{From } M: b = -(a + c) = 1/2.$$

$$-a - 3(1/2) + 2c = 0 \Rightarrow -a + 2c = 3/2.$$

$$\text{Adding with } a + c = -1/2: 3c = 1 \Rightarrow c = 1/3.$$

$$a = -1/2 - 1/3 = -5/6.$$

$$\text{Thus, } a = -5/6, b = 1/2, c = 1/3.$$

Sol 3. Correct Option: (a)

The argument of sine ($\omega t - kx$) must be dimensionless:

$$[\omega t] = [M^0 L^0 T^0] \Rightarrow [\omega] = [T^{-1}]$$

$$[kx] = [M^0 L^0 T^0] \Rightarrow [k] = [L^{-1}]$$

Ratio:

$$\left[\frac{\omega}{k} \right] = \left[\frac{T^{-1}}{L^{-1}} \right] = [L^1 T^{-1}] = [\text{Velocity}]$$

Sol 4. Correct Option: (b)

$$\text{Surface tension } [T] = [M^1 L^0 T^{-2}].$$

$$\text{Let } T = E^a v^b F^c:$$

$$[M^1 L^0 T^{-2}] = [ML^2T^{-2}]^a [LT^{-1}]^b [MLT^{-2}]^c$$

$$M : a + c = 1$$

$$L : 2a + b + c = 0 \Rightarrow a + b + (a + c) = 0 \Rightarrow$$

$$a + b + 1 = 0$$

$$T : -2a - b - 2c = -2 \Rightarrow 2(a + c) + b = 2 \Rightarrow$$

$$2(1) + b = 2 \Rightarrow b = 0$$

$$\text{From } L: a + 0 + 1 = 0 \Rightarrow a = -1.$$

$$c = 1 - a = 1 - (-1) = 2.$$

$$\text{Thus, } [T] = [E^{-1} v^0 F^2].$$

Sol 5. Correct Option: (a)

$$v \propto \lambda^a g^b \rho^c$$

$$[LT^{-1}] = [L]^a [LT^{-2}]^b [ML^{-3}]^c$$

$$M : c = 0$$

$$T : -2b = -1 \Rightarrow b = 1/2$$

$$L : a + b - 3c = 1 \Rightarrow a + 1/2 = 1 \Rightarrow a = 1/2$$

$$\text{Thus, } a = 1/2, b = 1/2, c = 0.$$

Sol 6. Correct Option: (d)

- (a), (b), (c) are dimensionally consistent (s_n is distance per 1 second, with dimension $[LT^{-1}]$ matching RHS).
- For de-Broglie wavelength:

$$[\lambda] = [L], \quad \left[\frac{h}{mv^2} \right] = \frac{[ML^2T^{-1}]}{[M][L^2T^{-2}]} = [T] \neq [L]$$

Hence (d) is dimensionally inconsistent (correct formula is $\lambda = h/mv$).

Sol 7. Correct Option: (c)

The exponent must be dimensionless:

$$\left[\frac{x^2}{\alpha k_B T} \right] = 1 \Rightarrow [\alpha] = \frac{[x^2]}{[k_B T]} = [M^{-1} T^2]$$

From $F = \alpha \beta$:

$$[\beta] = \frac{[F]}{[\alpha]} = \frac{[M^1 L^1 T^{-2}]}{[M^{-1} L^2 T^0]} = [M^1 L^{-1} T^{-2}]$$

Sol 8. Correct Option: (a)

$$\text{Let } L \propto h^a G^b c^c.$$

$$[L^1] = [ML^2T^{-1}]^a [M^{-1}L^3T^{-2}]^b [LT^{-1}]^c$$

$$M : a - b = 0 \Rightarrow a = b$$

$$L : 2a + 3b + c = 1 \Rightarrow 5a + c = 1$$

$$T : -a - 2b - c = 0 \Rightarrow -3a - c = 0 \Rightarrow c = -3a$$

$$5a - 3a = 1 \Rightarrow 2a = 1 \Rightarrow a = 1/2, b =$$

$$1/2, c = -3/2.$$

$$\text{Thus, } [L] = [h^{1/2} G^{1/2} c^{-3/2}].$$

Sol 9. Correct Option: (a)

$$[E] = [ML^2T^{-2}].$$

$$\text{Let } E = p^a A^b t^c:$$

$$[ML^2T^{-2}] = [MLT^{-1}]^a [L^2]^b [T]^c$$

$$M : a = 1$$

$$L : a + 2b = 2 \Rightarrow 1 + 2b = 2 \Rightarrow b = 1/2$$

$$T : -a + c = -2 \Rightarrow -1 + c = -2 \Rightarrow c = -1$$

$$\text{Thus, } [E] = [p^1 A^{1/2} t^{-1}].$$

Sol 10. Correct Option: (a)

$$[v_c] = [LT^{-1}].$$

Testing $\frac{\eta}{\rho D}$:

$$\left[\frac{\eta}{\rho D} \right] = \frac{[ML^{-1}T^{-1}]}{[ML^{-3}][L]} = \frac{[ML^{-1}T^{-1}]}{[ML^{-2}]} = [LT^{-1}]$$

Matching velocity dimensions. R_e is the dimensionless Reynolds number.

Sol 11. Correct Option: (c)

The exponent argument must be dimensionless:

$$\left[\frac{\alpha V}{R\theta} \right] = [M^0 L^0 T^0]$$

Since $[R\theta] = [\text{Energy}] = [M^1 L^2 T^{-2}]$ and $[P][V] = [\text{Energy}]$, we have $[R\theta] = [P][V]$.

Thus, $[\alpha][V]/([P][V]) = 1 \Rightarrow [\alpha] = [P]$ or matching volume exponent $[\alpha V] = [M^0 L^0 T^0] \Rightarrow [\alpha] = [L^{-3}]$.

Sol 12. Correct Option: (a)

Let $m_P \propto h^a c^b G^c$.

$$[M^1] = [ML^2 T^{-1}]^a [LT^{-1}]^b [M^{-1} L^3 T^{-2}]^c$$

$$M : a - c = 1$$

$$L : 2a + b + 3c = 0$$

$$T : -a - b - 2c = 0 \Rightarrow a + b + 2c = 0$$

Adding L and T equations:

$$(2a + b + 3c) - (a + b + 2c) = 0 \Rightarrow a + c = 0 \Rightarrow c = -a.$$

$$\text{Substitute in } M: a - (-a) = 1 \Rightarrow 2a = 1 \Rightarrow a = 1/2, c = -1/2.$$

$$b = -a - 2c = -1/2 - 2(-1/2) = 1/2.$$

$$\text{Thus, } m_P = [h^{1/2} c^{1/2} G^{-1/2}].$$

Sol 13. Correct Option: (a)

$$\left[\frac{e^2}{\varepsilon_0} \right] = [Fr^2] = [ML^3 T^{-2}].$$

$$[hc] = [ML^2 T^{-1}][LT^{-1}] = [ML^3 T^{-2}].$$

$$\text{Ratio: } \frac{[ML^3 T^{-2}]}{[ML^3 T^{-2}]} = [M^0 L^0 T^0].$$

Sol 14. Correct Option: (a)

$$[Q] = [L^3 T^{-1}], [P/l] = [ML^{-2} T^{-2}], [\eta] = [ML^{-1} T^{-1}], [r] = [L].$$

Poiseuille's formula gives:

$$Q = \frac{\pi P r^4}{8 \eta l} = \frac{\pi}{8} \eta^{-1} r^4 \left(\frac{P}{l} \right)^1$$

$$\text{Thus, } a = -1, b = 4, c = 1.$$

Sol 15. Correct Option: (c)

Resistance has dimensions $[M^1 L^2 T^{-3} A^{-2}]$, which requires an electrical fundamental quantity (like A or Q). Pure mechanical bases $\{F, v, \rho\}$ lack any electric dimension.

Sol 16. Correct Option: (a)

The argument of the exponential function must be dimensionless:

$$[\alpha t^2] = 1 \Rightarrow [\alpha] = \frac{1}{[t^2]} = [T^{-2}] = [M^0 L^0 T^{-2}]$$

Sol 17. Correct Option: (a)

$$\nu \propto L^a F^b \mu^c$$

$$[T^{-1}] = [L]^a [MLT^{-2}]^b [ML^{-1}]^c$$

$$M : b + c = 0 \Rightarrow c = -b$$

$$T : -2b = -1 \Rightarrow b = 1/2 \Rightarrow c = -1/2$$

$$L : a + b - c = 0 \Rightarrow a + 1/2 - (-1/2) = 0 \Rightarrow a = -1$$

$$\text{Thus, } a = -1, b = 1/2, c = -1/2.$$

Sol 18. Correct Option: (a)

$$\text{From } [x] = [a] \Rightarrow [a] = [L].$$

$$\text{From dimensionless sine argument: } [2\pi vt/b] = 1 \Rightarrow [b] = [vt] = [LT^{-1}][T] = [L].$$

$$\text{From dimensionless cosine argument: } [2\pi x/c] = 1 \Rightarrow [c] = [x] = [L].$$

Therefore:

$$\left[\frac{ab}{c} \right] = \frac{[L][L]}{[L]} = [L] = [M^0 L^1 T^0]$$

Sol 19. Correct Option: (a)

$$[F] = [m]^a [v]^b [r]^c$$

$$[M^1 L^1 T^{-2}] = [M]^a [LT^{-1}]^b [L]^c$$

$$M : a = 1$$

$$T : -b = -2 \Rightarrow b = 2$$

$$L : b + c = 1 \Rightarrow 2 + c = 1 \Rightarrow c = -1$$

$$\text{Thus, } F = k \frac{mv^2}{r}.$$

Sol 20. Correct Option: (d)

Dimensional analysis cannot determine numerical constants, fails for non-algebraic functions, and cannot uniquely resolve relations involving more variables than independent base dimensions.

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